



Research Article

Open Access (CC-BY-SA)

Integrated E-Learning System Development Using Next.js with Anti-Cheating Testing and Payment Integration

Eva Yumami^{1*}; Muhammad Ikhsan Wibowo²; Agusviyanda³; M. Khairul Anam⁴

¹ Politeknik Negeri Bengkalis, Jl. Bathin Alam, Sungai Alam, Bengkalis, Riau, 28711, Indonesia

² Institut Kesehatan Payung Negeri Pekanbaru, Jl. Tamtama No.6, Payung Sekaki, Pekanbaru, Riau, 28292, Indonesia

³ Universitas Samudra, Jl. Prof Syarif Thayeb, Meurandeh, Langsa City, Aceh, 24416, Indonesia

⁴ evayumami@polbeng.ac.id; ² mikhsanw@polbeng.ac.id; ³ agusviyanda27@gmail.com; ⁴ khairulanam@unsam.ac.id

* Corresponding author

Article history: Received April 30, 2026; Revised May 02, 2026; Accepted May 04, 2026; Available online May 07, 2026

Abstract

The advancement of information technology has significantly increased the adoption of e-learning in the educational process. However, many e-learning systems still face limitations, such as the lack of integration between learning management, evaluation systems, and payment features. This study aims to design and implement a web-based e-learning system called NexusLearn that integrates course management, Computer-Based Testing (CBT) with anti-cheating mechanisms, and a payment gateway into a unified platform. The system is developed using the Software Development Life Cycle (SDLC) method, which includes requirement analysis, system design, implementation, and testing. The application is built using the Next.js framework and applies Role-Based Access Control (RBAC) to manage user access based on roles. The results show that the system successfully integrates learning, assessment, and transaction features into a single platform. The CBT module supports dynamic question navigation, real-time timers, automatic answer saving, and anti-cheating mechanisms such as tab-switch detection and automatic submission. The payment feature enables secure and automated transactions through Midtrans integration. Based on Black-box Testing results, all system functionalities operate as intended across key features, including authentication, course management, examination, and payment processing. These findings indicate that the NexusLearn system effectively improves integration, security, and efficiency in e-learning environments and has strong potential for implementation in educational institutions.

Keywords: E-learning, Computer Based Test, Next.js, Payment Gateway, Role-Based Access Control.

Introduction

The advancement of information and communication technology has significantly transformed the educational paradigm, particularly with the increasing adoption of e-learning. E-learning has become a primary solution for providing flexible and accessible education without spatial and temporal limitations [1], [2], [3]. In addition, e-learning systems support integrated learning processes, including content delivery, user interaction, and evaluation within a single digital platform [4].

However, existing e-learning platforms still face several limitations, such as the lack of integration between learning management, evaluation systems, and payment features, as well as suboptimal user experience [5]. In addition, maintaining academic integrity in online examinations remains a critical challenge due to the high potential for cheating. Therefore, the implementation of Computer-Based Testing (CBT) combined with anti-cheating mechanisms and testing strategies is required to ensure fairness, reliability, and system robustness [6].

Based on these problems, this study aims to develop an integrated web-based e-learning system called NexusLearn using the Next.js framework. The system integrates course management, CBT with anti-cheating mechanisms, and payment features into a unified platform. The development process follows a structured approach to ensure systematic design, implementation, and testing. Furthermore, the application of the Learning Management System (LMS) concept enables more effective and organized learning management [7].

This study contributes by providing an integrated e-learning solution that combines learning management, secure CBT implementation, and payment integration within a single system. The proposed system is expected to support educational institutions in managing learning, evaluation, and transactions efficiently and securely.

Method

This study employs the Software Development Life Cycle (SDLC) method in developing the NexusLearn e-learning system. This method consists of several stages, namely requirement analysis, system design, implementation, and testing. Each stage is carried out systematically to ensure that the developed system meets user

requirements [8]. These stages are part of the SDLC approach, which is applied in a structured manner in system development [9]. The stages of the SDLC software development method are illustrated in Figure 1.

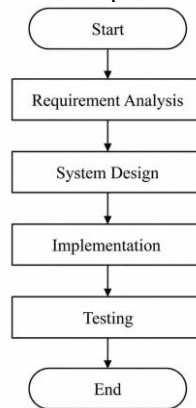


Figure 1. Method SDLC

The figure shows the system development flow starting from requirement analysis to testing, which is carried out sequentially to ensure that each process runs properly and produces a system that meets user requirements. In addition, the figure indicates that each stage is interconnected, where the output of one phase becomes the input for the next phase. This structured flow helps ensure that the system is developed systematically, minimizes inconsistencies, and improves the overall reliability of the resulting application.

A. Requirement Analysis

The research begins with the requirement analysis phase, which aims to identify both functional and non-functional requirements of the NexusLearn e-learning system to be developed. The analysis process is conducted through literature review, observation of existing e-learning systems, and identification of user needs based on roles (Admin, Instructor, and Student). The identified functional requirements include user management, course and learning material management, implementation of Computer-Based Testing (CBT) with automatic grading, and the application of anti-cheating mechanisms. In addition, the system must support integration with a payment gateway to facilitate online course transactions.

Meanwhile, non-functional requirements include system performance in terms of responsiveness, user data security, cross-device compatibility, and system scalability to accommodate increasing numbers of users. The results of this analysis phase serve as the foundation for the system design stage.

B. System Design

• Use Case

System modeling using a use case diagram is employed to describe the interaction between actors and the system in a structured manner, thereby facilitating the system design and implementation process [10]. The diagram illustrates the main functionalities available in the system as well as the access rights of each actor, namely Admin, Instructor, and Student. Through the use case diagram, the overall service flow of the system can be understood, ranging from user and course management, the implementation of Computer-Based Testing (CBT), to the payment process.

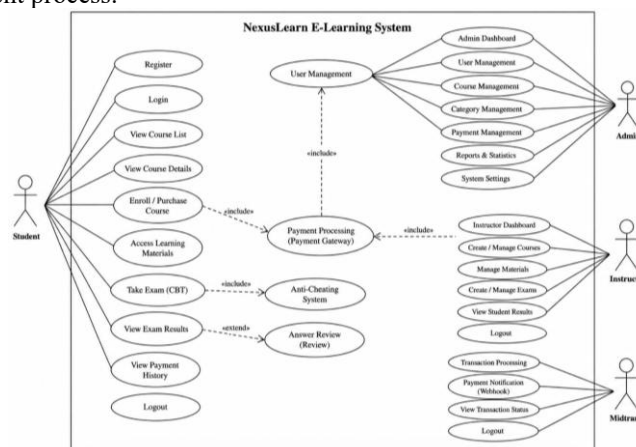


Figure 2. Use Case

• System Architecture

System architecture is used to describe the structure and relationships among components within the NexusLearn e-learning system. The purpose of this architectural design is to provide an overview of how data flows and interactions between modules operate in an integrated manner. The NexusLearn e-learning system

is designed using a modular monolithic architecture pattern based on the Next.js framework to improve performance and system efficiency [11] in which the frontend and backend are integrated into a single project based on the Next.js framework [12], [13]. This approach enables more structured and efficient development, while also simplifying system management and maintenance processes.

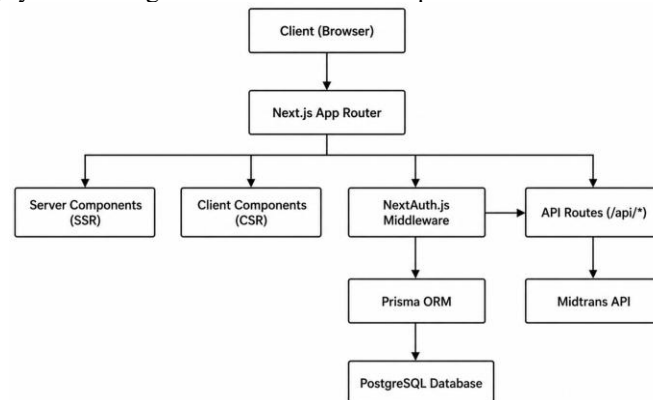


Figure 3. System Architecture

- Database Design

The database design stage is carried out to define the data structure and relationships among entities within the NexusLearn e-learning system. The main entities include User, which is related to Enrollment, Course, Payment, Progress, and Exam Attempt. The Course entity is associated with Material and Exam, and is also linked to Category for course classification. This design aims to ensure that data management is well-structured and supports all system processes effectively [14]. The designed database structure is illustrated in the following figure, as shown in Figure 4.

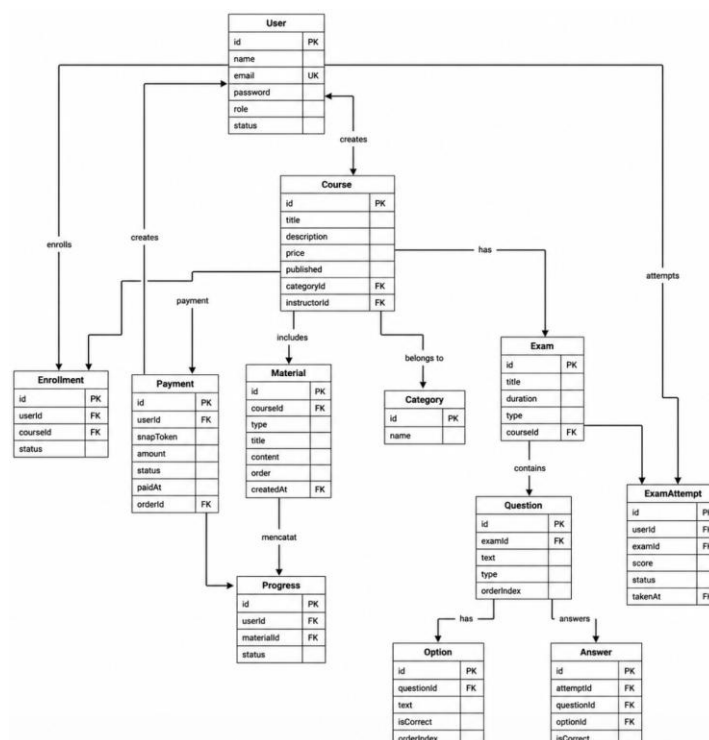


Figure 4. Database e-learning

- Payment Gateway Integration Design

The payment gateway integration is designed to support transaction processes within the NexusLearn e-learning system. The integration utilizes the Midtrans Snap service, which provides various payment methods in an automated and secure manner ([15], [16]). The payment flow involves interactions between users, the frontend, backend API, Midtrans services, and the database. This process includes transaction creation, payment token generation, and receiving webhook notifications to update the payment status in real time. Additionally, this integration ensures transaction security through signature validation and payment status verification provided by Midtrans. The designed integration flow is illustrated in the following figure.

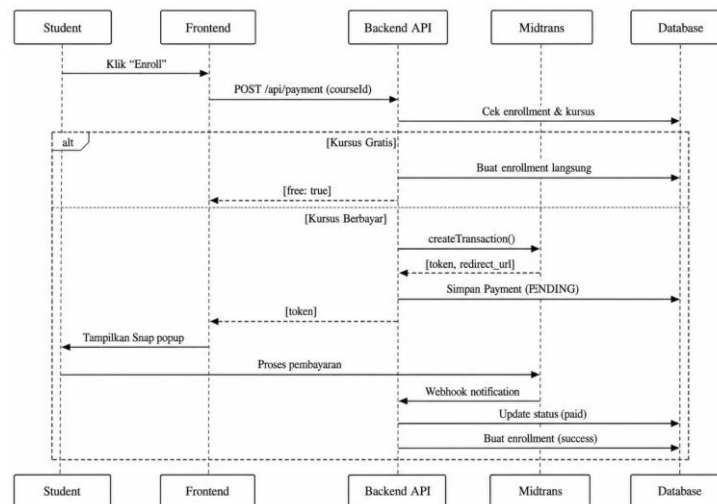


Figure 5. Payment Gateway Integration Design.

C. System Implementation

The implementation stage is a continuation of the design phase within the SDLC, where the designed system is developed into a web-based application. The result of this study is the implementation of the NexusLearn e-learning system as a web-based platform using the Next.js framework, which integrates course management, Computer-Based Testing (CBT), and payment mechanisms into a unified system. The developed system applies the Role-Based Access Control (RBAC) concept [1], ensuring that each user has different access rights based on their roles [14], [17], namely Admin, Instructor, and Student. The system provides an authentication feature that directs users to dashboards according to their respective roles. The Admin dashboard displays aggregated data such as the number of users, courses, students, and enrollments, along with system management features. The Instructor dashboard presents course information, the number of students, and exams, as well as features for creating and managing courses. Meanwhile, the Student dashboard displays enrolled courses, learning progress, and exam results, as illustrated in Figure.

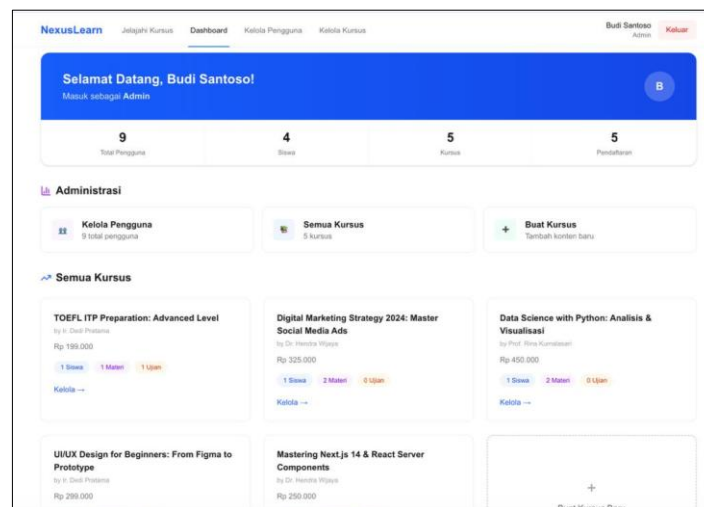


Figure 6. Admin Dashboard

Figure 6 shows the Admin Dashboard of the NexusLearn system, which provides aggregated information such as the number of users, courses, students, and enrollments. This result demonstrates that the system successfully integrates data management into a single interface, enabling administrators to monitor system activities efficiently. The presence of centralized control features indicates that the system supports effective management and aligns with the concept of an integrated Learning Management System (LMS). This also confirms that the implementation meets the functional requirements identified in the analysis phase.

The course management feature allows users to add courses, manage learning materials, and create examinations. The materials are available in various formats, including text, video, and PDF, and the system records users' learning progress. This indicates that the system fulfills the characteristics of an integrated Learning Management System (LMS) [18]. The CBT system is implemented with dynamic question displays, navigation between questions, and a real-time timer [19]. User answers are automatically saved during the examination to ensure data consistency, as illustrated in the following figure 7.

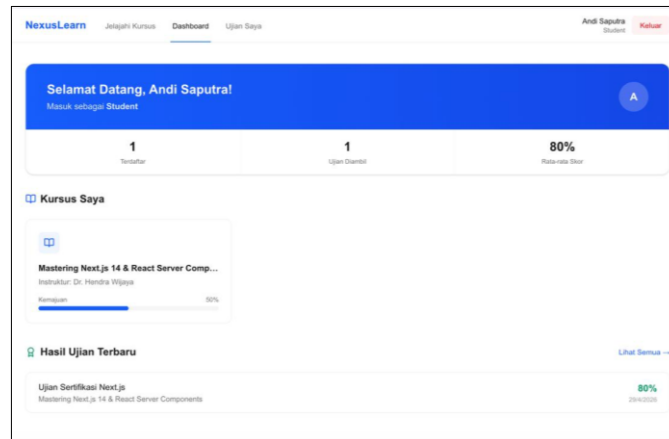


Figure 7. Student Dashboard

Figure 7 presents the Student Dashboard, which displays enrolled courses, learning progress, and recent exam results. This result is consistent with the previously described functionality of the student dashboard. The visualization of progress and results indicates that the system is capable of tracking user learning activities in real time [20]. This feature enhances user engagement and provides immediate feedback, which is essential in improving learning effectiveness. Therefore, the system not only delivers content but also supports continuous monitoring of student performance.

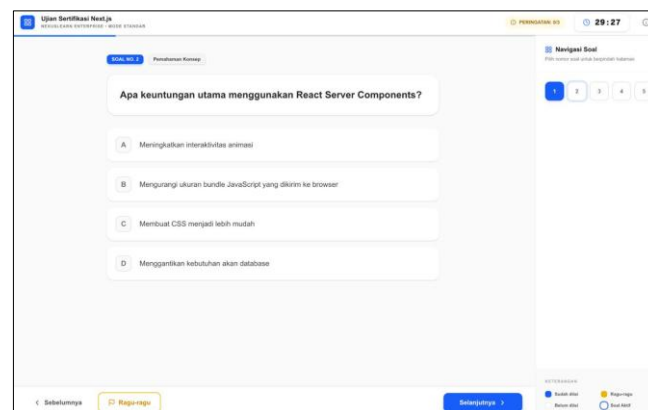


Figure 8. System CBT

Figure 8 illustrates the implementation of the Computer-Based Test (CBT) interface, including dynamic questions, navigation between questions, and a real-time timer. This result shows that the system supports a structured and controlled examination process. During the examination, user answers are automatically saved to ensure data consistency and prevent data loss. This indicates that the system is reliable and suitable for conducting online assessments [21].

After the examination is completed, the system automatically calculates and displays the results, including the final score, number of correct answers, pass/fail status, completion time, and violation logs. The system also provides answer reviews as feedback to support the learning process. To maintain exam integrity, the system is equipped with anti-cheating mechanisms [19]. These mechanisms include browser tab-switch detection, which triggers warnings to the user. If the number of violations exceeds the predefined limit, the system will automatically submit the exam.

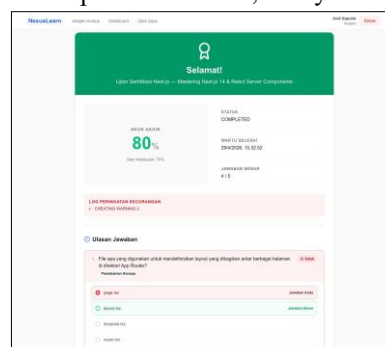


Figure 9. CBT Results

Figure 9 shows the CBT result interface, which includes the final score, number of correct answers, pass/fail status, completion time, and violation logs. After the examination is completed, the system automatically calculates and displays the results. The system also provides answer reviews as feedback to support the learning process.

To maintain exam integrity, the system implements anti-cheating mechanisms such as browser tab-switch detection and warning systems. If the number of violations exceeds the predefined limit, the system automatically submits the exam. This feature enhances the reliability of the system in preventing cheating and ensuring fair evaluation [21].

The following presents the implementation of the payment feature in the e-learning system, which is integrated with the Midtrans payment gateway. This feature enables users to purchase courses directly through various available payment methods, making the course enrollment process more flexible, secure, and automated [22]. This implementation is illustrated in Figure 10.

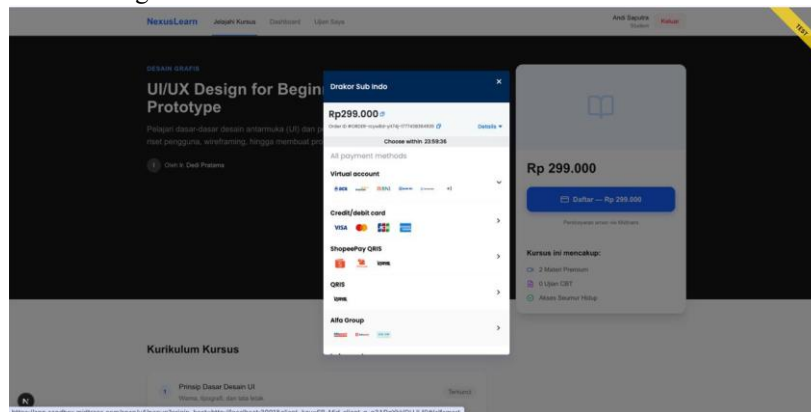


Figure 10. Payment Feature

Figure 10 illustrates the implementation of the payment feature integrated with the Midtrans payment gateway. This result shows that the system successfully supports multiple payment methods and automates the transaction process. The integration ensures secure and real-time transaction handling, which enhances user convenience and system efficiency. This confirms that the system is capable of supporting monetization features within an e-learning platform

D. System Testing

The testing phase in the SDLC is conducted to ensure that the system operates according to the requirements identified in the earlier stages. System testing is performed using the Black-box Testing method to verify that each function of the NexusLearn e-learning system works as expected [8]. This method focuses on testing system functionality without considering the internal structure of the code, by providing specific inputs and observing the resulting outputs. Testing is carried out on key features such as authentication, course management, CBT implementation, and payment gateway integration. The test results are then compared with the expected outputs to ensure that the system functions properly and meets the specified requirements, as presented in the following table.

Table 1. Black-Box Testing

No	Test Scenario	Input	Expected Output	Result
1	New user registration	Valid data (name, email, password)	Account created, redirected to login	Success
2	Login with valid credentials	Correct email and password	Redirected to dashboard	Success
3	Login with invalid credentials	Incorrect password	Error message displayed	Success
4	Access admin page by Student	URL /admin/*	Redirected to dashboard	Success
5	View course catalog	Access /courses	Course list displayed	Success
6	Free course enrollment	Click enroll on free course	Enrollment successful, access granted	Success
7	Paid course payment	Click enroll, process via Midtrans	Snap popup displayed, payment processed	Success
8	Start CBT exam	Click Start Exam, enter fullscreen	Exam started, timer running	Success

No	Test Scenario	Input	Expected Output	Result
9	Tab-switch detection during exam	Switch browser tab	Warning appears, counter increases	Success
10	Auto-submit after 3 violations	Switch tab 3 times	Exam automatically submitted	Success
11	Manual exam submission	Answer questions, click Submit	Score displayed	Success
12	Auto-submit when time expires	Let timer run out	Exam automatically submitted	Success
13	Learning progress tracking	Open and complete material	Progress bar updated	Success
14	Midtrans webhook	Successful payment notification	Status PAID, auto-enrollment	Success
15	Invalid webhook signature validation	Invalid signature	Response 403 Forbidden	Success

Table 1 presents the results of Black-box Testing conducted on key system functionalities. All test scenarios show successful results, indicating that the system operates according to the specified requirements. This result confirms that the NexusLearn system is functionally reliable and meets both functional and non-functional requirements identified in earlier stages.

Conclusion

Based on the results of this study, the NexusLearn e-learning system has been successfully developed as an integrated web-based platform using the Next.js framework, which combines course management, Computer-Based Testing (CBT), anti-cheating mechanisms, and payment gateway integration into a single system. The implementation results demonstrate that the system is capable of supporting structured learning processes, secure online examinations through anti-cheating features such as tab-switch detection and automatic submission, as well as automated and secure transaction management via payment gateway integration. The application of Role-Based Access Control (RBAC) further ensures appropriate access control for each user role.

Furthermore, the results of Black-box Testing indicate that all system functionalities operate as expected, confirming the reliability and effectiveness of the system in handling learning, evaluation, and transaction processes. These findings highlight that the integration of Next.js, CBT with anti-cheating mechanisms, and payment systems significantly improves system efficiency, security, and usability.

Therefore, NexusLearn has strong potential to be implemented in educational institutions as a comprehensive and integrated e-learning solution. Future development may include the addition of advanced features such as video conferencing, discussion forums, and AI-based proctoring to further enhance learning quality and monitoring capabilities.

References

- [1] N. Yudhi Handika, W. Fitriani, P. Studi Sistem Komputer, F. Sains dan Teknologi, U. Pembangunan Panca Budi, and C. Author, "Perancangan Sistem E-Learning Kursus Online Berbasis Web Menggunakan Laravel," *Jurnal Nasional Teknologi Komputer*, vol. 5, no. 3, 2025.
- [2] M. Armansyah and A. Sekti, "Pengembangan Fitur Penelitian Tugas Real-Time Pada Sistem E-Learning Berbasis Web Untuk Meningkatkan Interaktivitas Siswa," *Bulletin Of Computer Science Research*, vol. 6, no. 1, pp. 325–331, Dec. 2025, doi: 10.47065/bulletincsr.v6i1.891.
- [3] M. A. Almaiah, A. Al-Khasawneh, and A. Althunibat, "Exploring the critical challenges and factors influencing the E-learning system usage during COVID-19 pandemic," *Educ. Inf. Technol. (Dordr.)*, vol. 25, no. 6, pp. 5261–5280, Nov. 2020, doi: 10.1007/s10639-020-10219-y.
- [4] E. Susanti, S. Assegaff, M. Sistem Informasi, U. Dinamika Bangsa, and J. JI Jend Sudirman Thehok-Jambi, "Sistem E-Learning Berbasis Web Pada SMA Negeri 1 Merangin," 2022.
- [5] A. Salem Mohamedhen, N. Arfaoui, R. Ejballi, and F. Nanne, "Learning Object From Emergence To Nowadays: Systematics Literature Review," *J. Theor. Appl. Inf. Technol.*, vol. 29, no. 4, 2024, [Online]. Available: <https://ieeexplore.ieee.org>
- [6] Kartarina, O. Travelian, Moch. Syahrir, N. Islamiah, and wahyu Kamil Syarifaturrahman, "Fisher-Yates Shuffle and Linear Congruent Algorithm in the Mini Challenge of Computational Thinking Task," *International Journal of Informatics and Computation (IJICOM)*, vol. 7, 2025, doi: 10.35842/ijicom.

-
- [7] R. Novita *et al.*, “Implementasi The Concurrent Development Model Untuk Membangun Learning Management System,” *JURNAL INOVTEK POLBENG*, vol. 9, no. 1, p. 2024, 2024.
- [8] A. S. Panyili, Sukardi, and E. Alfonsius, “Pengembangan Learning Management System (LMS) Berbasis Website Untuk Meningkatkan Efektivitas Dan Efisiensi Pembelajaran,” *Riau Jurnal Teknik Informatika*, vol. 4, no. 1, Mar. 2024, doi: 10.30606/rjti.v4i1.3254.
- [9] M. D. Agisfio, “Pengembangan Sistem E-Learning Berbasis Web Di SMA XYZ,” *Jurnal Informatika dan Teknik Elektro Terapan*, vol. 14, no. 1, Jan. 2026, doi: 10.23960/jitet.v14i1.8213.
- [10] S. Syarifah and K. Abd Latif, “Perancangan E-Learning Berbasis Web Di SMK Sidratul Munthaha Yapis Wamena Papua Dengan Metode Blackbox,” *Rabit : Jurnal Teknologi dan Sistem Informasi Univrab*, vol. 9, no. 2, pp. 343–354, Jul. 2024, doi: 10.36341/rabit.v9i2.4701.
- [11] A. Rakhmansyah, A. Jabbar, M. Ramzy, and W. Haryono, “Development Digital Payment Application Using Next.js and Xendit with Prototyping-Agile Approach,” *Jurnal Ilmiah Informatika dan Komputer (Informatech)*, vol. 2, no. 1, pp. 9–19, 2025, doi: 10.69533.
- [12] M. Lutfi Maftuh and Nurkhamid, “Implementasi Next.js Dan Headless Cms Untuk Peningkatan Kualitas Website Mi Al Islam Pare,” Feb. 2024. doi: 10.21831/jiety.v1i2.203.
- [13] D. Wahyu Widarti and M. Khamal Akbar, “Peningkatan Performa Aplikasi Web SIAKAD Menggunakan Next.js Di Puskordat Stimata,” Jan. 2026. doi: 10.24076/joism.2026v7i2.2453.
- [14] H. Romadhan and Mansur, “Sistem Informasi Pembayaran Pendidikan berbasis Web menggunakan Keamanan Role-based Access Control,” *Sistemasi: Jurnal Sistem Informasi*, vol. 15, no. 2, pp. 614–627, Dec. 2025, doi: 10.32520/stmsi.v15i2.5920.
- [15] R. A. Siahaan and R. A. Sianturi, “Analisis Perbandingan Payment Gateway untuk Sistem Pembayaran Berbasis Aplikasi dengan Comparative Study,” *Jurnal Teknologi Informasi dan Ilmu Komputer*, vol. 11, no. 2, pp. 291–296, Apr. 2024, doi: 10.25126/jtiik.20241127680.
- [16] Robby Cokro Buwono, Rikie Kartadie, and Muhammad Kurnia Ramadhan, “Perancangan dan implementasi sistem payment gateway MIDTRANS untuk UMKM Batik Lurik,” *INFOTECH: Jurnal Informatika & Teknologi*, vol. 5, no. 2, pp. 228–238, Dec. 2024, doi: 10.37373/infotech.v5i2.1334.
- [17] M. Firmansyah, “Implementasi Role-Based Access Control Pada Sistem Informasi Layanan Administrasi Pada Lembaga Bahasa Um Metro,” *JMIK (Jurnal Mahasiswa Ilmu KomPUTER)*, vol. 6, no. 2, pp. 113–122, 2025, doi: 10.24127/ilmukomputer.v6i2.9875.
- [18] F. Mahabul¹, M. Subhan², O. I. Pramadita³, A. Fahriza⁴, and A. Ekabudi⁵, “Pemanfaatan Learning Management System (LMS) Untuk Meningkatkan Efektifitas Pembelajaran,” *Jurnal Teknologi Pendidikan Dan Pembelajaran (JTTP)*, vol. 03, no. 01, pp. 27–34.
- [19] R. Setiawan, “Optimasi Pengalaman Pengguna Dan Prototyping Untuk Penilaian Otomatis Dan Pencegahan Kecurangan,” *bit-Tech*, vol. 7, no. 2, pp. 299–306, Dec. 2024, doi: 10.32877/bt.v7i2.1758.
- [20] D. Ramadhan, Z. Sitorus, and M. S. Penjaitan, “Design and Development of an SDLC-Based E-Learning Application as a Learning Medium Using the Blended Learning Model,” *Bulletin of Information Technology (BIT)*, vol. 5, no. 3, pp. 173–179, 2024, doi: 10.47065/bit.v5i2.1480.
- [21] S. One Janah, E. Fatma Sari, R. Hermawan, R. Putra Ramadhan, and A. Ardiansyah, “Website Sipusat (Sistem Informasi Penilaian Ujian Sekolah Anti-Cheating) Sebagai Solusi Membantu Para Guru Dalam Melakukan Penilaian UTS dan UAS,” *Jurnal Sistem Informasi Akuntansi*, vol. 03, no. 02, pp. 37–43, Oct. 2023, doi: 10.31294/jasika.v3i2.2950.
- [22] H. Fitriawan, B. Sundari, and Mardiana, “Komunikasi host to host pada payment gateway dengan private network dan token akses,” *Prosiding Seminar Nasional Ilmu Teknik Dan Aplikasi Industri Fakultas Teknik Universitas Lampung*, pp. 277–283, Nov. 2022, doi: 10.23960/prosidingsinta.v5i.92.
-